Plan-to-Clone: Planning-Driven Data Distribution Design for Contact-Rich Manipulation

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***Abstract*— Learning robust policies for contact-rich manipulation remains challenging due to the difficulty of collecting demonstrations that adequately capture complex contact transitions and recovery behaviors. Existing imitation learning methods often rely on human teleoperation, which is costly, difficult to scale, and provides limited coverage of rare but critical corrective behaviors. We propose a planner-in-the-loop demonstration synthesis framework that autonomously generates long-horizon, contact-rich manipulation trajectories without teleoperation. Our method couples a global contact-implicit planner for high-level task reasoning with a contact-implicit model predictive controller that enforces dynamic feasibility and systematically injects disturbances during execution. This integration not only ensures physical consistency, but also enables controlled shaping of the training data distribution by generating structured recovery behaviors. In real-world experiments, the distilled policies execute from proprioceptive, visual, and tactile observations, without motion tracking required by direct planner execution. Through simulated and real-world experiments on simple maze, spinner, box pushing and rotating, bimanual lifting, balancing, and whole-body manipulation, we demonstrate that the proposed framework produces feasible and diverse datasets that yield policies capable of robust closed-loop contact interaction. Controlled ablations further show that recovery data improves policy robustness while it expands the effectively demonstrated state coverage; once coverage saturates, additional recovery data can displace nominal supervision and degrade performance, underscoring data-distribution design in contact-rich imitation learning.**

Index Terms— Behavior Cloning, Demonstration Synthesis, Contact-Rich Manipulation, Planning and Control Integration

I. Introduction

Robotic manipulation involving dynamic and contact-rich interactions poses significant challenges due to hybrid dynamics, mode switches, and sensitivity to disturbances. Behavior cloning has emerged as a promising solution for learning robust manipulation behaviors by modeling complex, multimodal behavior distributions [2]. However, the effectiveness of behavior cloning critically depends on the coverage and structure of the demonstration dataset. In practice, demonstrations are often collected via human teleoperation, which is labor-intensive, difficult to scale, and rarely covers critical recovery behaviors.

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To address these limitations, we propose a planner-in-the-loop demonstration synthesis framework (Fig. 1) tailored for behavior cloning that systematically generates structured supervision with minimal human intervention. Our method combines a global contact-implicit planner for high-level task reasoning with a local contact-implicit model predictive controller (CI-MPC) for trajectory stabilization and reactive correction. The global planner [23] provides long-horizon task-level structure, while CI-MPC [14] enforces dynamic feasibility during execution and enables controlled disturbance injection to generate recovery behaviors.

A key advantage of this planning-driven approach is the systematic generation of recovery behaviors. By initializing unstable states or injecting controlled disturbances, CI-MPC produces physically consistent corrective trajectories that are difficult to elicit consistently through teleoperation. These trajectories expose the learner to near-failure states that are rarely represented in conventional demonstration datasets.

Why distill the planner? Although the planning pipeline can execute the tasks considered here, we use it as a development-time expert rather than as the intended deployment controller. Distillation (i) replaces online object-state feedback, task-specific models, and tuned solvers with a visuomotor policy operating from proprioceptive, visual, and tactile observations; (ii) permits repeated offline planning attempts while retaining only successful nominal and recovery executions, so the expert need not be sufficiently reliable for direct deployment; and (iii) amortizes the planning and data-generation effort across policy executions while converting task-specific solutions into a common observation-action format that supports scalable co-training across tasks and data sources. This expert-at-development-time, policy-at-deployment structure follows prior work in guided policy search, controller cloning, and planning-based skill distillation [9, 16, 26]. Section IV-D quantifies this on a simulated task where the distilled policy outperforms the planner under matched state-estimation noise, though not under delay.

Our contributions in this work are threefold:

- We integrate contact-implicit methods spanning global planning, local optimization, and MPC to execute long-horizon contact-rich tasks from high-level goals and systematically generate recovery behaviors.
- We show that successful planner-generated behaviors can be distilled into observation-based visuomotor policies that reproduce robust contact interactions without requiring the planning pipeline at deployment.
- We introduce a structured data-distribution design perspective for contact-rich behavior cloning and show em-

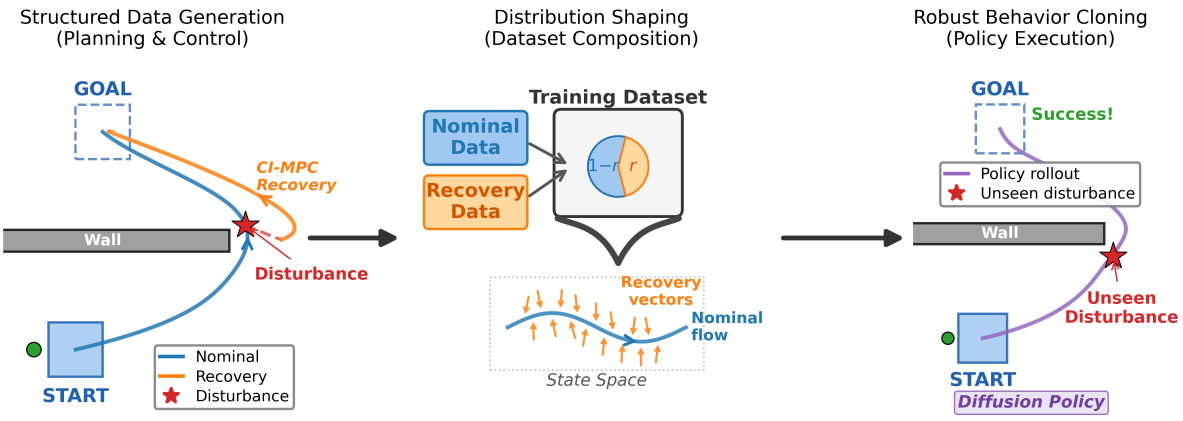


Fig. 1: Overview of the proposed framework. **(a)** GQDP, CI-TO, and CI-MPC generate nominal and disturbance-induced recovery trajectories. **(b)** Mixing these trajectories at ratio r shapes the training distribution; recovery transitions steer perturbed states toward the nominal flow. **(c)** The cloned policy reproduces these corrections under unseen disturbances.

pirically that recovery data improves policy robustness while expanding demonstrated state coverage, but can degrade performance by displacing nominal supervision once coverage saturates.

This capability is particularly valuable for whole-body manipulation, where coordinated contacts across the hands, forearms, and torso make it difficult to collect nominal and recovery demonstrations reliably with conventional end-effector teleoperation interfaces.

II. Related Work

We categorize methods for scaling demonstration collection and improving policy robustness into four complementary directions.

A. Physical Model-Based Demonstration Generation

Physical model-based methods leverage known physical properties, kinematic and dynamic constraints, and explicit formulations such as Dynamic Movement Primitives (DMPs) [10], Guided Policy Search (GPS) [16], and Tube MPC-based cloning [26]. These approaches provide structured and feasible supervision from system models and constraints; GPS and Tube MPC-based cloning also exemplify the use of a model-based expert during development and a learned policy at deployment. However, they primarily focus on nominal task execution rather than systematically generating recovery behaviors under disturbances or unexpected contacts.

B. Planning-Based Imitation Learning

Planning-based imitation learning uses planners to generate or refine demonstrations for policy learning, often improving generalization and performance in contact-rich or long-horizon tasks. Notable approaches include Monte Carlo Tree Search-based EfficientImitate [30], diffusion-based latent planners [6], and sampling-based planners such as GLIDE [18], DIAL-MPC [28], and Jacta [11].

Recent work also explores physics-driven data generation through planning [29]. Related work [1] uses examples generated by a Global Quasi-Dynamics Planner (GQDP) [23] to guide reinforcement learning for contact-rich whole-body manipulation. Our method likewise begins with GQDP, followed by contact-implicit trajectory optimization (CI-TO), implemented using Inverse Dynamics Trajectory Optimization (IDTO) [14], and contact-implicit model predictive control (CI-MPC). In contrast, we use this planning-and-control pipeline to directly generate behavior-cloning demonstrations and deliberately shape their nominal/recovery distribution through disturbance injection.

C. Demonstration Data Generation and Augmentation

Data generation and augmentation approaches aim to efficiently construct extensive and diverse datasets for robust imitation learning. Techniques such as MimicGen [22], SkillGen [8], DexMimicGen [11], Universal Manipulation Interface (UMI) for large-scale human-generated data collection [3], and SCIZOR for self-supervised data curation [31] show scalable demonstration synthesis, aggregation, and curation pipelines. EgoMimic [12] leverages egocentric human video with synchronized hand-tracking to construct large-scale datasets co-trained with robot data, while language-based task specification further expands task diversity [9]. Generated expert data has also been used for behavior cloning in active scene recognition [19], while RACER [4] augments demonstrations with language-guided failure-recovery trajectories. In contrast, we generate physically grounded recovery data through closed-loop reactive control.

D. Sim-and-Real Co-Training

Sim-and-real co-training approaches combine simulated and real-world data to bridge the simulation-to-reality gap and enhance policy robustness. Notable examples include the "Sim-and-Real Co-Training" framework by Maddukuri et al. [21], which highlights the importance of data composition and visual discrepancies in enhancing policy performance. Wei et al. [27] empirically analyze policies for planar pushing

tasks, showing substantial gains through combined training and visual domain variation. Liu et al. [20] proposed a consensus-based deep reinforcement learning approach, showing enhanced performance and consistency in pick-and-place tasks. These works focus primarily on domain transfer and cross-domain dataset composition. In contrast, our approach emphasizes structured demonstration synthesis within simulation and hardware, using physically consistent planning and control to improve robustness to contact dynamics and disturbances.

Collectively, prior work demonstrates the value of model-based planning, scalable data generation, and dataset composition strategies for imitation learning in contact-rich manipulation. Our work builds upon these directions by framing planning and control not only as execution tools, but as mechanisms for shaping the training data distribution through systematic generation of recovery behaviors.

III. METHOD

Our method systematically generates demonstration trajectories through a structured pipeline consisting of global planning, trajectory refinement, and local reactive control. Global planning provides a coarse task-level solution, trajectory refinement ensures kinematic and dynamic feasibility, and local reactive control adapts execution to disturbances and unexpected contact transitions.

A. Global Quasi-Dynamic Planning

Initially, a global trajectory is computed via GQDP [23] using the initial states of the actuated robot $\mathbf{q}_0^a$, the initial state of the underactuated object $\mathbf{q}_0^{\mathbf{u}}$, and the goal state of the underactuated object $\mathbf{q}_{\mathbf{goal}}^{\mathbf{u}}$. GQDP employs a quasi-dynamic contact model to discover a coarse path to reach the goal state. The output is a sequence of robot states $\mathbf{q}^{\mathbf{a}}$, object states $\mathbf{q}^{\mathbf{u}}$, and control actions $\mathbf{a}$ that is not necessarily feasible.

B. Kinematic Trajectory Optimization

The output of GQDP $(\mathbf{q}^{\mathbf{a}}, \mathbf{q}^{\mathbf{u}}, \mathbf{a})$ is refined through Kinematic Trajectory Optimization (TrajOpt), ensuring kinematic feasibility. This step explicitly accounts for joint limits and collision avoidance, providing a mechanically consistent intermediate trajectory suitable for dynamic refinement.

C. Contact-Implicit Trajectory Optimization

The kinematically feasible trajectory from Kinematic TrajOpt is further refined using CI-TO, implemented using IDTO [14]. CI-TO accounts for dynamic constraints and contact interactions, producing dynamically feasible trajectories for execution.

D. Contact-Implicit Model Predictive Control

The resulting trajectory from CI-TO serves as the input to CI-MPC. CI-MPC operates in real time, adapting trajectories to disturbances and unanticipated contacts; thus capturing rich reactive behaviors essential for robust policy learning. The output of CI-MPC is a set of reactive trajectories represented by sensor readings I_{camera} (for image-based policies) and states $(\mathbf{q}^{\mathbf{a}}, \mathbf{a})$ for training state-based policies.

E. Demonstration Collection

The state trajectories (and/or images) generated through this sequential process (i.e., global planning, kinematic optimization, dynamic refinement, and reactive control) constitute a behavior cloning dataset. Moreover, the diversity and composition of the dataset can be controlled through domain randomization and disturbance injection, enabling systematic generation of recovery behaviors.

Algorithm 1 Demonstration Generation Pipeline

Require: $\mathbf{q}_0^{\mathbf{a}}, \mathbf{q}_0^{\mathbf{u}}, \mathbf{q}_{\mathbf{goal}}^{\mathbf{u}}$
Ensure: Demonstration dataset $(\mathbf{I}_{\mathbf{camera}}, \mathbf{q}^{\mathbf{a}}, \mathbf{a})$
  Generate initial trajectory with GQDP:
    $(\mathbf{q}^{\mathbf{a}}, \mathbf{q}^{\mathbf{u}}, \mathbf{a}) \leftarrow \text{GQDP}(\mathbf{q}_0^{\mathbf{a}}, \mathbf{q}_0^{\mathbf{u}}, \mathbf{q}_{\mathbf{goal}}^{\mathbf{u}})$
  Refine trajectory with Kinematic TrajOpt:
    $(\mathbf{q}^{\mathbf{a}}, \mathbf{q}^{\mathbf{u}}, \mathbf{a}) \leftarrow \text{KinematicTrajOpt}(\mathbf{q}^{\mathbf{a}}, \mathbf{q}^{\mathbf{u}}, \mathbf{a})$
  Dynamically refine trajectory with CI-TO:
    $(\mathbf{q}^{\mathbf{a}}, \mathbf{q}^{\mathbf{u}}, \mathbf{a}) \leftarrow \text{CI-TO}(\mathbf{q}^{\mathbf{a}}, \mathbf{q}^{\mathbf{u}}, \mathbf{a})$
  Real-time execution with CI-MPC:
    $(\mathbf{I}_{\mathbf{camera}}, \mathbf{q}^{\mathbf{a}}, \mathbf{a}) \leftarrow \text{CI-MPC}(\mathbf{q}^{\mathbf{a}}, \mathbf{q}^{\mathbf{u}}, \mathbf{a})$
  Collect dataset $(\mathbf{I}_{\mathbf{camera}}, \mathbf{q}^{\mathbf{a}}, \mathbf{a})$ for training

IV. EXPERIMENTS

We conducted several experiments to evaluate the effectiveness of the proposed approach across various scenarios, emphasizing the integration of CI-TO and CI-MPC with GQDP-generated trajectories. Specifically, our experiments are designed to answer the following research questions:

- **RQ1**: Can CI-TO and CI-MPC refine and stabilize long-horizon plans from GQDP? (Section IV-A)
- **RQ2**: Can CI-TO stabilize dynamically infeasible trajectories from GQDP? (Section IV-B)
- **RQ3**: When does adding recovery data to the training set improve policy robustness, and when does it hurt? (Section IV-C)
- **RQ4**: How does distillation affect robustness to state-estimation noise and delay relative to CI-MPC? (Section IV-D)
- **RQ5**: Can behavior-cloned policies reproduce the recovery behaviors of CI-MPC? (Section IV-E)
- **RQ6**: Can the framework generate effective demonstrations for whole-body contact-rich manipulation? (Section IV-F)

Table I summarizes the experimental tasks used to address these questions. Since real-world evaluation is difficult to standardize and scale, we use simulation for controlled quantitative comparisons and hardware experiments to demonstrate physical feasibility, consistent with recent scalable robot-policy evaluation practice [17].

A. Integration of Long-Horizon Behavior from GQDP

We first evaluate the ability of CI-TO and CI-MPC to refine and execute long-horizon behaviors generated by GQDP. We design a simple maze environment in simulation to highlight the need for global reasoning and local stabilization.

TABLE I: Summary of experimental tasks and corresponding research questions.

Task	Domain	Description	RQ
Simple Maze	Sim	Planar pushing through a maze (sphere pushes 1.0 m box)	1, 3
Spinner	Sim	Rotating a free spinner to a goal angle with two fingers	3, 4
Jaco Box Push and Rotate	Sim	Pushing and rotating a box to a goal with Jaco arm	3
Bimanual Lift	Real	Lifting a box 0.3 m with Punyo-1 dual-arm robot	2
Box Balancing	Real	Pivoting a 15 cm cube to 45 deg roll and balancing	5
Whole-Body Rotation	Real	Rotating a cardboard box via whole-body manipulation	6

The Simple Maze experiment involves pushing a 1.0 m × 1.0 m box from the near side to the far side of a maze using a spherical pusher with a radius of 0.1 m. Collisions are defined among the box, the sphere, and the maze walls, requiring the pusher to move the box while avoiding obstacles, as illustrated in Fig. 1a.

First, we obtain a global trajectory where the sphere pushes the box from the start to the goal position from GQDP. The resulting trajectory serves as the initial solution for CI-TO. Subsequently, we employ the optimized output as the reference for CI-MPC. This experiment evaluates whether CI-MPC can track, stabilize, and refine long-horizon (and potentially infeasible) behaviors generated by GQDP.

CI-MPC reduced the box-center path length from 8.72 m to 6.22 m (28.6%) and the final position error from 0.262 m to 0.009 m. GQDP produced 113 steps, of which the first 50 were used as the CI-TO horizon. During CI-MPC execution, the nominal state was updated every 0.1 s. This task is inherently challenging for local CI-TO alone, highlighting the need to integrate global planning with local optimization and reactive control.

B. Stabilization of Dynamically Infeasible Trajectories

A major limitation of GQDP is that it creates segments in the trajectory that are stable independently, but the inter-segments are not guaranteed to be stable. Hence, it is not guaranteed to generate dynamically stable trajectories. To evaluate whether CI-TO can stabilize such unstable trajectories, we design a task that involves lifting a box by 0.3 m using a bimanual robot, Punyo-1 [7][1]. We use an OptiTrack motion-capture system to track the manipuland.

Before CI-TO, several inter-segment transitions exhibit separation between the end effectors and the box. After CI-TO, contact is maintained at every time step, indicating that the trajectory has been stabilized. Figure 2 shows a rollout of the resulting trajectory.

The contact breaks in the GQDP trajectory occur at inter-segment transitions. CI-TO, on the other hand, enforces both kinematic and dynamic contact constraints to compose these segments into a physically coherent motion.

[1]Punyo-1 is composed of two Kinova Jaco Gen-2 arms, each with soft visuotactile sensors [13] as end effectors and an array of 7 air-filled pressure-based sensors that cover each arm.

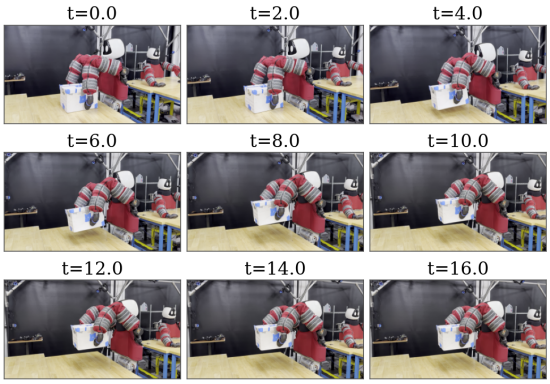
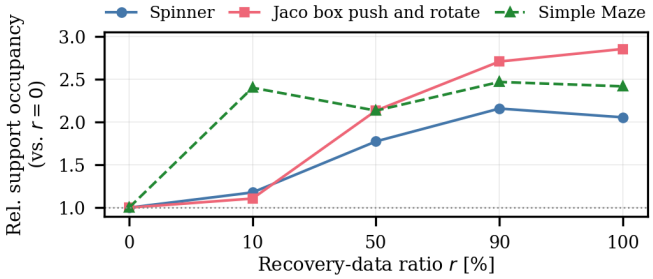
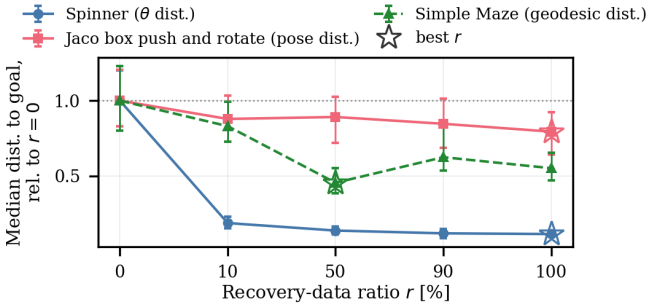


Fig. 2: Snapshots from the bimanual box lifting experiment.



(a) Breadth of the demonstrated support, relative to r=0.



(b) Median distance to the goal under disturbed evaluation, relative to r=0 (lower is better; stars mark each task's best ratio). Absolute r=0 medians: Spinner 0.80 rad, Jaco 0.13, maze 3.60 m.

Fig. 3: Effect of the recovery-data ratio r across three tasks (bootstrap 95% CIs; Spinner/Jaco: pooled disturbed conditions, n=225; maze: small disturbances, n=20–21).

C. Ablation Study on Data Composition

The pipeline gives explicit control over the composition of the training data. We use it to ask a question that teleoperated data collection cannot easily answer: *when does adding recovery demonstrations help, and can it hurt?* We define the mixing ratio r as the fraction of recovery trajectories ($\mathcal{D}_{\text{rec}}$) in the dataset, the remaining $1-r$ being nominal trajectories ($\mathcal{D}_{\text{nom}}$), and study $r \in \{0, 0.1, 0.5, 0.9, 1.0\}$ on three tasks of increasing horizon and contact complexity.

1) Tasks and Protocol: Two *short-horizon* tasks use a 2–4 s CI-MPC planning horizon and a single sustained contact

interaction: *Spinner*, where two actuated fingers rotate a free spinner to a goal angle, with recovery data generated from a widened initial-state distribution; and *Jaco box push and rotate*, where a 7-DoF arm pushes a 15 cm box to a goal pose while rolling it 90°, with recovery data generated by repeated pushes applied to the box during execution. The third task is the *Simple Maze* of Section IV-A: a *long-horizon* task (5 s planning horizon seeded by a GQDP global plan) in which the box must be driven around a wall through dramatic contact transitions—engaging, releasing, and re-approaching the object. All tasks share one protocol: 200 success-filtered demonstrations per dataset, one diffusion policy per dataset, and evaluation under four disturbance conditions (none/S/M/L) with 20–75 trials each. The support-occupancy measure of Fig. 3 counts grid cells of a fixed task-space region visited by ≥ 2 demonstrations (Spinner: $(\theta, \dot{\theta})$, 0.1 grid; Jaco: box (x, y), 1 cm; maze: pre-wall box $(x, y) \in [0, 1.5] \times [-0.5, 0.5]$ m, 5 cm, equal-size subsampling); the ≥ 2 threshold discounts cells touched by a single stray trajectory. We report the median task-space distance to the goal: the angular distance $|\theta_{\text{err}}|$ for Spinner; a pose distance for Jaco,[2] and for the maze the *geodesic* (free-space shortest-path) distance, computed by Dijkstra on a 2.5 cm occupancy grid with the wall inflated by the box half-extents (Euclidean distance mis-orders progress before the wall is rounded), with bootstrap confidence intervals and pairwise Mann–Whitney U tests with Holm correction (Table II).

2) Short-Horizon Tasks: Broader Coverage Improves Robustness: Figure 3 shows the two short-horizon tasks. Each recovery demonstration traverses novel states before rejoining the nominal solution, so the demonstrated support broadens with r (up to 2.1× for Spinner and 2.8× for Jaco; Fig. 3a), and the distance to the goal under disturbances falls accordingly (Fig. 3b): the median drops to 11% of the r=0 value on Spinner ($0.80 \rightarrow 0.09$ rad; Mann–Whitney $p = 1.1 \times 10^{-40}$) and to 79% on Jaco ($0.128 \rightarrow 0.102$; $p = 3.3 \times 10^{-3}$; the decrease is not strictly monotonic). Neither task shows an interior optimum, and even $r = 1.0$ leaves undisturbed performance essentially intact (Spinner $p = 0.29$; Jaco median 0.029 versus 0.038, $p = 0.04$)—recovery trajectories pass through the nominal corridor and therefore largely subsume its supervision.

3) Long-Horizon Maze Task: Higher Diversity Does Not Necessarily Help: The maze task behaves differently. Its demonstrations are already highly diverse—the global plan admits many routes around the wall—so recovery data expands the demonstrated support only up to a point: the occupied region grows steeply at small r and then saturates (Fig. 3a), after which additional recovery data only *replaces* nominal supervision on largely shared states. Correspondingly, performance is non-monotonic (Fig. 3b; Table II): $r = 50\%$ achieves the lowest median geodesic distance under every evaluation condition (1.58–1.75 m versus 3.40–3.97 m for $r = 0\%$) and is the sole member of the best statistical group under three of four conditions, significantly outperforming $r = 0\%$ (adjusted $p \leq 3.7 \times 10^{-4}$) and $r = 100\%$ (adjusted $p \leq 0.022$) under *all* conditions; only its advantage over $r = 90\%$ under medium disturbances is marginal (adjusted $p = 0.09$). Moreover, $r = 100\%$ degrades even the undisturbed case (median 2.00 m versus 1.58 m, adjusted $p = 3.6 \times 10^{-4}$).

The disturbance-*magnitude* axis on the same task shows that severity itself is not the driver (Fig. 4): pre-wall occupancy grows monotonically with severity (12% to 34%), yet performance does not follow it. Under small-disturbance evaluation, the medium and mixed-magnitude datasets form the best statistical group; the all-magnitude mixture reaches a median geodesic distance of 1.98 m versus 3.60 m for the no-disturbance baseline (Holm-adjusted $p = 3.3 \times 10^{-6}$). The small-only and large-only datasets fall in the weaker group (small-only versus baseline, adjusted $p = 1.0$). Broader mixtures of useful recovery conditions therefore help, but severity alone is not monotonically beneficial.

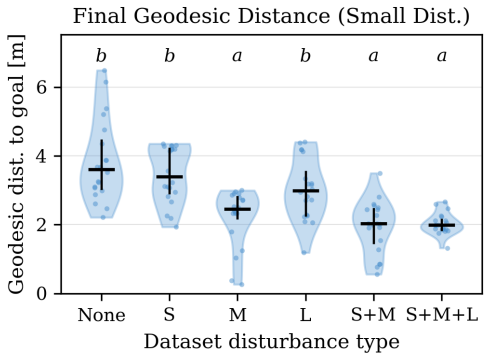


Fig. 4: Maze task: geodesic distance to goal for policies trained with different disturbance magnitudes, evaluated under small disturbances (n=20 trials per condition). Letters denote statistically homogeneous groups (Mann–Whitney U, Holm-corrected, $\alpha = 0.05$).

4) Distribution Support vs. Ratio: Taken together, the three tasks support a simple reading: *the recovery-data ratio is not itself the governing variable. What matters is whether additional recovery demonstrations expand useful state coverage or merely replace nominal supervision after coverage has saturated.* As long as recovery data keeps expanding the useful demonstrated support, as in the short-horizon tasks at every r, adding more of it keeps reducing the distance to the goal, with no observed penalty. Once the support saturates, as on the maze beyond $r \approx 0.5$, further recovery data can displace nominal supervision, and performance degrades on both disturbed and nominal evaluations. This saturation is *consistent with* the interior optimum at $r = 50\%$, though we do not claim it causally explains it. In our study, the post-saturation regime arises only on the maze; on Spinner and Jaco, the support never saturates within $r \leq 1$, so the degradation regime does not occur there.

[2] $\sqrt{\|\Delta p\|^2 + \|\Delta q\|^2}$ with quaternion chord distance $\|\Delta q\|$, weighting position and orientation equally, as in the CI-MPC cost.

TABLE II: Maze task: geodesic distance to goal [m] by training ratio r: median [bootstrap 95% CI] over n=20–21 trials. Superscript letters denote statistically homogeneous groups per column (Mann–Whitney U, Holm-corrected, $\alpha = 0.05$); 'a' marks the best group.

Training data	Eval: none	Eval: small	Eval: medium	Eval: large
0%	3.40 [3.04, 3.66]^{e}	3.60 [3.08, 4.11]^{e}	3.84 [3.33, 4.24]^{c}	3.97 [3.50, 4.13]^{c}
10%	2.92 [2.90, 2.93]^{d}	2.99 [2.96, 3.17]^{d}	3.30 [3.13, 3.39]^{b}	3.34 [3.13, 3.50]^{b}
50%	**1.58 [1.53, 1.65]**^{a}	**1.64 [1.52, 1.80]**^{a}	**1.75 [1.58, 2.02]**^{a}	**1.68 [1.48, 1.94]**^{a}
90%	2.49 [2.26, 2.68]^{c}	2.25 [2.09, 2.89]^{c}	2.26 [1.94, 3.54]^{a}	3.85 [3.56, 4.06]^{c}
100%	2.00 [1.88, 2.02]^{b}	1.98 [1.87, 2.12]^{b}	2.75 [1.93, 3.20]^{b}	2.58 [2.07, 3.41]^{b}

D. Planner vs. Policy under Degraded State Estimation

Why is a cloned policy needed when the CI-MPC expert already solves the task? Beyond removing the deployment-time need for object models and instrumented state estimation, distillation can *exceed* the expert under identical sensing. We quantify this on the Spinner task of Section IV-C, comparing the CI-MPC demonstrator with the diffusion policies distilled from it ($r = 0$ and $r = 0.5$). We corrupt the state fed to the entire control stack of each controller identically—zero-mean Gaussian noise (σ_q on positions, $\sigma_v = 10\sigma_q$ on velocities, resampled per control tick) or a pure observation delay—while the simulated ground truth remains untouched. Each condition uses 25–50 episodes with paired initial states.

Figure 5 shows the results; since initial states are paired across controllers and conditions, we use exact McNemar tests with Holm correction across the eight corruption levels. The outcome is a robustness *trade-off*: the policy is stronger under matched state-estimation noise, while CI-MPC tolerates delay better. On clean observations, the policy substantially outperforms its own demonstrator (88% versus 36% success for $r = 0.5$; adjusted $p = 0.011$): it is trained only on the *successful* subset of demonstrations, so distillation filters out the expert's own failures. Under observation noise, the policies degrade gracefully and remain ahead at every level, although per-level differences are not significant at $n = 25$ (e.g., 36% versus 8% at $\sigma_q = 0.03$ rad, adjusted $p = 0.77$). The planner itself is non-monotonic: a small amount of noise *raises* its success rate from 36% to 74% ($p = 1.3 \times 10^{-3}$); we hypothesize that dither perturbs the warm start of the single-iteration solver out of a local minimum, an effect specific to the optimization loop and absent in the policies. Under delay the ranking reverses: 200 Hz replanning partially absorbs stale observations while the 10 Hz open-loop action chunks cannot (differences at 0.1–0.2 s not significant); all methods achieve low success rates beyond 0.1 s (planner solver divergences, 60 of 275 episodes, are counted as failures).

E. Dynamic Recovery Behaviors

In this experiment, we consider a real-world task where a 15 cm cube is pivoted along an edge to 45 deg roll and balanced using a bimanual humanoid robot. We evaluate whether the learned policy can reproduce the controller's dynamic recovery behavior, as shown in Fig. 6a.

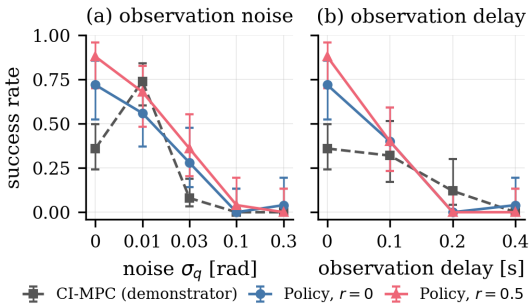


Fig. 5: Success rates of the CI-MPC demonstrator and distilled diffusion policies on the spinner task under (a) observation noise and (b) observation delay ($\sigma_v = 10\sigma_q$; 25–50 episodes per condition with paired initial states; error bars are 95% Wilson intervals).

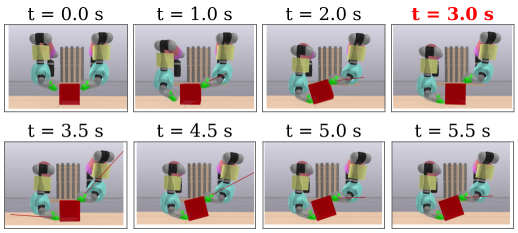
We collected 20 demonstrations of the CI-MPC balancing the box and trained a policy using these demonstrations to examine whether it can imitate the recovery behavior of the CI-MPC. The observation space included the robot's joint space, visuo-tactile sensor data from the palms (grayscale images), and two scene cameras. Figure 6b shows the policy reproducing the corrective behavior demonstrated by CI-MPC.

F. Whole-Body Contact-Rich Manipulation

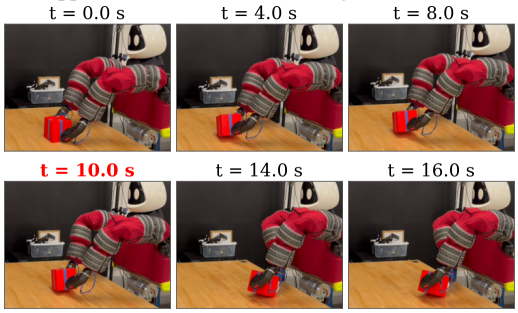
Lastly, we apply the framework to whole-body contact-rich rotation of a $0.41 \times 0.3 \times 0.26$ m cardboard box, requiring coordinated hand, forearm, and torso contacts that are difficult to demonstrate with conventional end-effector teleoperation [1].

Using the proposed framework, we collected 20 demonstrations on the hardware. For each trial, the box was slightly repositioned by a few centimeters to increase dataset diversity. To prevent excessive internal forces during robot-object interaction that could cause overcurrent in the actuators, we employed a compliant controller, as in [1].

For behavior cloning, we kept only trajectories in which the box was successfully rotated by 90 degrees. The observations comprise the robot's joint space, visuo-tactile signals from the palms, and two scene cameras; motion-capture data was used only for success detection. In the rollout of Fig. 7, at approximately 10 s the robot momentarily loses stability and nearly drops the box, but regains control and completes



(a) Recovery behavior from CI-MPC. The controller detects the box tipping and applies corrective forces to regain balance.



(b) The behavior-cloned policy reproduces the same corrective response learned from CI-MPC demonstrations.

Fig. 6: Snapshot of box balancing task.

the rotation—learned recovery behavior in a whole-body task.

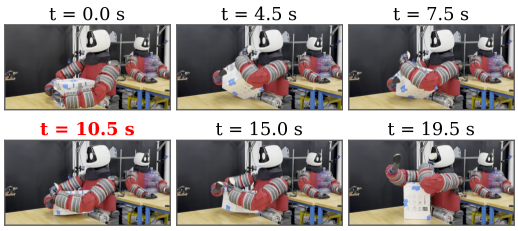


Fig. 7: Key frames from a rollout: at $t \approx 10.5\,\mathrm{s}$, the robot momentarily loses stability but recovers and completes the 90° rotation.

V. Discussion

We use model predictive control not only for execution, but also to generate structured behavior-cloning data. The composition ablations provide controlled quantitative evidence that robustness depends on training-distribution design, while hardware rollouts establish physical transfer of planner-generated behaviors.

A. The Role of Dataset Composition

The maze result assigns complementary roles to the two data sources: nominal trajectories define the intended task structure, while recovery trajectories supply the local corrective dynamics around it. Once the demonstrated support saturates, the two compete for probability mass in the learned model. This offers a possible explanation for the interior optimum in Section IV-C and aligns with observations in generative modeling, where mixtures of clean and diverse noisy samples are essential for robust manifolds [5]. Diversity alone is therefore not the objective; the composition of nominal and corrective data shapes closed-loop behavior.

B. Relation to Covariate Shift

Behavior cloning suffers from covariate shift: small errors move the policy beyond the expert-induced state distribution [24]. DAgger queries the expert on learner-visited states [25]; DART perturbs demonstrations offline [15]. Our approach likewise expands off-nominal supervision, but through controlled physical disturbances corrected by CI-MPC before training, avoiding rollouts of a partially trained policy on hardware. The disturbance distribution is, however, fixed in advance and may not match the learned policy's own error distribution; the support-saturation analysis above indicates when this trade-off stops paying off.

C. Robustness Trade-offs of Distillation

Section IV-D shows that success-conditioned distillation can outperform its own CI-MPC demonstrator. The improvement is not uniform, however. The distilled policy tolerates state-estimation noise better than the planner, while observation delay affects it more, since stale observations violate the timing assumed by its open-loop action chunks. Distillation thus changes, rather than uniformly improves, the robustness profile, and the dominant degradation mode of the target sensing stack should guide the choice of controller.

VI. Conclusion

We presented a framework for automated demonstration synthesis of long-horizon, contact-rich manipulation without reliance on human teleoperation. By coupling a global contact-implicit planner with reactive contact-implicit MPC, the system generates physically consistent trajectories that include structured recovery behaviors under controlled disturbances.

We validated the approach through simulated and real-world experiments, ranging from the simple maze to whole-body manipulation with an upper-body humanoid robot. CI-MPC can leverage global plans as priors, stabilizing and refining them when locally infeasible. Controlled ablations further show that recovery data improves policy robustness while it expands the demonstrated state coverage, and can degrade performance by displacing nominal supervision once coverage saturates.

More broadly, planning and control can serve as development-time experts that generate and shape training data, while behavior cloning distills their contact-rich capabilities into visuomotor policies that need no planning stack at deployment.

VII. Limitations and Future Work

The proposed approach is limited to structured tasks that can be modeled with sufficient accuracy and computational tractability; in particular:

- IDTO currently relies on simple contact geometries (half-spaces, spheres), requiring a manual approximation of more complex ones.

- Configuring GQDP, CI-TO, and CI-MPC required approximately 4–8 expert hours per task, dominated by contact-geometry approximation and tuning of contact geometry, friction, contact stiffness, and smoothing parameters.
- The framework depends on model fidelity; contact-model or dynamics errors degrade the generated demonstrations and hence the learned policy—it would correspondingly benefit from advances in simulation and planning through contact.

Beyond these constraints, a key open direction is predicting the coverage-saturation point—where additional recovery data stops helping—from task structure, disturbance characteristics, and policy architecture.

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REFERENCES

[1] Jose A. Barreiros, Aykut Özgün Önol, Mengchao Zhang, Sam Creasey, Aimee Goncalves, Andrew Beaulieu, Aditya Bhat, Kate M. Tsui, and Alex Alspach. Learning contact-rich whole-body manipulation with example-guided reinforcement learning. *Science Robotics*, 10(105):eads6790, 2025.

[2] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 2024.

[3] Cheng Chi, Zhenjia Xu, Chuer Pan, Eric Cousineau, Benjamin Burchfiel, Siyuan Feng, Russ Tedrake, and Shuran Song. Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots. In *Proceedings of Robotics: Science and Systems (RSS)*, 2024.

[4] Yinpei Dai, Jayjun Lee, Nima Fazeli, and Joyce Chai. Racer: Rich language-guided failure recovery policies for imitation learning. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 15657–15664. IEEE, 2025.

[5] Giannis Daras, Yeshwanth Cherapanamjeri, and Constantinos Costis Daskalakis. How much is a noisy image worth? data scaling laws for ambient diffusion. In *The Thirteenth International Conference on Learning Representations*, October 2024.

[6] Xinyu Geng et al. Latent diffusion planning for imitation learning. *arXiv preprint arXiv:2501.02364*, 2025.

[7] Aimee Goncalves, Naveen Kuppuswamy, Andrew Beaulieu, Avinash Uttamchandani, Katherine M. Tsui, and Alex Alspach. Punyo-1: Soft tactile-sensing upper-body robot for large object manipulation and physical human interaction. In *2022 IEEE 5th International Conference on Soft Robotics (RoboSoft)*, pages 844–851, 2022.

[8] Abhinav Goyal et al. Skillgen: Composing and generalizing manipulation skills from limited human demonstrations. *arXiv preprint arXiv:2401.09245*, 2024.

[9] Huy Ha, Pete Florence, and Shuran Song. Scaling up and distilling down: Language-guided robot skill acquisition. In Jie Tan, Marc Toussaint, and Kourosh Darvish, editors, *Proceedings of the 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 3766–3777. PMLR, Nov 2023.

[10] Auke Jan Ijspeert, Jun Nakanishi, and Stefan Schaal. Learning attractor landscapes for learning motor primitives. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2003.

[11] Zhenyu Jiang, Yuqi Xie, Kevin Lin, Zhenjia Xu, Weikang Wan, Ajay Mandlekar, Linxi Jim Fan, and Yuke Zhu. Dexmimicgen: Automated data generation for bimanual dexterous manipulation via imitation learning. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 16923–16930. IEEE, 2025.

[12] Simar Kareer, Dhruv Patel, Ryan Punamiya, Pranay Mathur, Shuo Cheng, Chen Wang, Judy Hoffman, and Danfei Xu. Egomimic: Scaling imitation learning via egocentric video, 2024.

[13] Naveen Kuppuswamy, Alex Alspach, Avinash Uttamchandani, Sam Creasey, Takuya Ikeda, and Russ Tedrake. Soft-bubble grippers for robust and perceptive manipulation. In *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 9917–9924, 2020.

[14] Vince Kurtz, Alejandro Castro, Aykut Özgün Önol, and Hai Lin. Inverse dynamics trajectory optimization for contact-implicit model predictive control. *The International Journal of Robotics Research*, 45(1):23–40, 2026.

[15] Michael Laskey, Jonathan Lee, Roy Fox, Anca Dragan, and Ken Goldberg. DART: Noise injection for robust imitation learning. In *Proceedings of the 1st Annual Conference on Robot Learning*, volume 78 of *Proceedings of Machine Learning Research*, pages 143–156. PMLR, 2017.

[16] Sergey Levine, Chelsea Finn, Trevor Darrell, and Pieter Abbeel. End-to-end training of deep visuomotor policies. *The Journal of Machine Learning Research*, 17:1334–1373, 2016.

[17] Xuanlin Li, Kyle Hsu, Jiayuan Gu, Oier Mees, Karl Pertsch, Homer Rich Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, and Ted Xiao. Evaluating real-world robot manipulation policies in simulation. In *Proceedings of the 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 3705–3728. PMLR, 2025.

[18] Xuanlin Li, Tong Zhao, Xinghao Zhu, Jiuguang Wang, Tao Pang, and Kuan Fang. Planning-guided diffusion policy learning for generalizable contact-rich bimanual manipulation, 2025.

[19] Shaopeng Liu, Chao Huang, and Hailong Huang. Behavior cloning-based active scene recognition via generated expert data with revision and prediction for domestic robots. *IEEE Transactions on Robotics*, 41:4180–4194, 2025.

[20] Wenxing Liu, Hanlin Niu, Wei Pan, Guido Herrmann, and Joaquin Carrasco. Sim-and-real reinforcement learning for manipulation: A consensus-based approach. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 3911–3917, 2023.

[21] Abhiram Maddukuri, Zhenyu Jiang, Lawrence Yunliang Chen, Soroush Nasiriany, Yuqi Xie, Yu Fang, Wenqi Huang, Zu Wang, Zhenjia Xu, Nikita Chernyadev, Scott Reed, Ken Goldberg, Ajay Mandlekar, Linxi Fan, and Yuke Zhu. Sim-and-real co-training: A simple recipe for vision-based robotic manipulation. In *Proceedings of Robotics: Science and Systems (RSS)*, Los Angeles, CA, USA, 2025.

[22] Ajay Mandlekar et al. Mimicgen: Scalable imitation learning via automated data generation. *Robotics: Science and Systems (RSS)*, 2023.

[23] Tao Pang, H. J. Terry Suh, Lujie Yang, and Russ Tedrake. Global planning for contact-rich manipulation via local smoothing of quasi-dynamic contact models. *Trans. Rob.*, 39(6):4691–4711, December 2023.

[24] Stephane Ross and Drew Bagnell. Efficient reductions for imitation learning. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, volume 9 of *Proceedings of Machine Learning Research*, pages 661–668. PMLR, 2010.

[25] Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pages 627–635. PMLR, 2011.

[26] Andrea Tagliabue and Jonathan P. How. Tube mpc-based behavioral cloning for robust zero-shot transfer. *IEEE Robotics and Automation Letters*, 2024.

[27] Yilun Wei et al. Empirical analysis of sim-and-real cotraining of diffusion policies for planar pushing, 2025.

[28] Haoru Xue, Chaoyi Pan, Zeji Yi, Guannan Qu, and Guanya Shi. Full-order sampling-based mpc for torque-level locomotion control via diffusion-style annealing, 2024.

[29] Lujie Yang, HJ Suh, Tong Zhao, Bernhard Paus Graesdal, Tarik Kelestemur, Jiuguang Wang, Tao Pang, and Russ Tedrake. Physics-driven data generation for contact-rich manipulation via trajectory optimization. *arXiv preprint arXiv:2502.20382*, 2025.

[30] Hang Yin et al. Efficientimitate: Planning with monte carlo tree search for efficient imitation learning. In *Conference on Robot Learning (CoRL)*, 2022.

[31] Yu Zhang, Yuqi Xie, Huihan Liu, Rutav Shah, Michael Wan, Linxi Fan, and Yuke Zhu. Scizor: A self-supervised approach to data curation for large-scale imitation learning, 2025.